

BCA (Semester-I) Examination, 2021**(Session : 2020-23)****COMPUTER APPLICATION****[Paper : BCA-101]****(Mathematical Foundation)**

Time : Three Hours

[Maximum Marks : 80]

Note: Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any five questions.

1. Find the eigen values and eigen vectors of the following matrix:

(a) $A = \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}$ (b) $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$

2. If $A = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 3 \\ -1 & 5 \end{bmatrix}$, $C = \begin{bmatrix} 3 & 5 \\ 1 & -2 \end{bmatrix}$,

find the value of x and y .

If $xA + yB = C$

3. Using matrix method, solve the following system of equations:

$$x + y + z = 6$$

$$2x + y - 3z = -5$$

$$3x - 2y + z = 2$$

4. If $y = (\tan^{-1} x)^2 (1 + x^2)^2 \cdot \frac{d^2 y}{dx^2} + 2x(1 + x^2) \cdot \frac{dy}{dx} = 2y$

prove that

$$\frac{1}{m} + y - \frac{1}{m} = 2x$$

$$\frac{d^2 y}{dx^2} + x \frac{dy}{dx} - m^2 y = 0$$

5. If $y = \frac{\sqrt{1-x^2}}{m}$, prove that $A = \begin{bmatrix} 1 & 2 \\ 12 & -1 \end{bmatrix}$

6. Find the maximum and minimum of the function:

$$f(x, y) = x^3 + y^3 - 12x - 3y + 15$$

7. Solve the following differential equations:

(a) $(x^2 - yx^2)dy + (y^2 + xy^2)dx = 0$

(b) $\frac{dy}{dx} = \sec(x + y)$

8. Evaluate the integrals:

(a) $\int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \sin(x + y) dy dx$

(b) $\int_0^2 \int_0^x (x^2 + y^2) dy dx$

9. Solve the following differential equations:

(a) $x\sqrt{1-y^2} dx + y\sqrt{1-x^2} dy = 0$

(b) $y dx - x dy = xy dx$

10. Solve the following:

$$\frac{d^3 y}{dx^3} - \frac{d^2 y}{dx^2} + \frac{dy}{dx} - y = 0$$

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